

Axler Exercise 3D.1: The Inverse of an Inverse

A Proof and Conceptual Commentary

Linear Algebra Done Right, 4th Edition

Context in the Text

This is Exercise 1 in Section 3D (*Invertibility and Isomorphic Vector Spaces*) of *Linear Algebra Done Right*, 4th edition. Section 3D opens by defining invertibility and the inverse of a linear map (Definition 3.59), establishing that inverses are unique (Theorem 3.60), and introducing the notation T^{-1} (3.61). This exercise is the first in the section and is intentionally elementary: it asks the reader to check that the defining equations of an inverse are symmetric in T and T^{-1} , and to invoke uniqueness. It is less a proof than a careful reading of the definition.

Prerequisites

- **Definition 3.59 (Invertible, inverse):** $T \in \mathcal{L}(V, W)$ is invertible if there exists $S \in \mathcal{L}(W, V)$ such that $ST = I_V$ and $TS = I_W$. Such an S is called an inverse of T .
- **Theorem 3.60 (Uniqueness of the inverse):** An invertible linear map has a unique inverse, denoted T^{-1} .
- **Notation 3.61:** If $T \in \mathcal{L}(V, W)$ is invertible, then T^{-1} is the unique element of $\mathcal{L}(W, V)$ satisfying $T^{-1}T = I_V$ and $TT^{-1} = I_W$.

Statement

Theorem (Exercise 3D.1). *Suppose $T \in \mathcal{L}(V, W)$ is invertible. Then T^{-1} is invertible and*

$$(T^{-1})^{-1} = T.$$

Proof

Since T is invertible, its inverse $T^{-1} \in \mathcal{L}(W, V)$ satisfies

$$T^{-1}T = I_V \quad \text{and} \quad TT^{-1} = I_W.$$

These same two equations, read from the perspective of T^{-1} , say precisely that $T \in \mathcal{L}(V, W)$ is an inverse of $T^{-1} \in \mathcal{L}(W, V)$: a map in $\mathcal{L}(V, W)$ that composes with T^{-1} on each side to give the appropriate identity. Therefore T^{-1} is invertible.

Since inverses are unique (Theorem 3.60), T must be *the* inverse of T^{-1} , i.e.,

$$(T^{-1})^{-1} = T. \quad \blacksquare$$

Key Concept

Key Concept. The core observation is that Definition 3.59 is *symmetric* in T and T^{-1} . The two conditions

$$T^{-1}T = I_V \quad \text{and} \quad TT^{-1} = I_W$$

make no distinction between which map is the “original” and which is the “inverse” — each is the inverse of the other. The only thing the proof needs to add beyond this symmetry observation is uniqueness (Theorem 3.60), to upgrade “ T is *an* inverse of T^{-1} ” to “ T is *the* inverse of T^{-1} ,” which is what the notation $(T^{-1})^{-1} = T$ asserts.

This same symmetry argument recurs throughout algebra: in group theory, the inverse of g^{-1} is g for exactly the same reason.